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Original Research

Compared to dual-energy X-ray absorptiometry, bioelectrical impedance effectively monitors longitudinal changes in body composition in children and adolescents with obesity during a lifestyle intervention

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ABSTRACT

Accurate and efficient body composition assessment is essential for diagnosing and monitoring childhood obesity in clinical and research settings. This study evaluated the agreement between bioelectrical impedance analysis (BIA) and dual-energy X-ray absorptiometry (DXA) in assessing body composition and its changes during a long-term lifestyle intervention in children and adolescents with obesity. We hypothesized that BIA is comparable to DXA in tracking longitudinal changes in body composition in this population. The study included 92 children and adolescents 5 to 17 years of age (mean 10.5 ± 2.9 years) classified as obese at the time of measurement (body mass index-standard deviation score ≥ 2 SD). From January 1, 2014, to December 31, 2017, they participated in a lifestyle intervention at Randers Regional Hospital. Body composition was assessed using BIA and DXA at baseline and follow-up (mean 20.5 ± 9.3 months). Bland–Altman plots, correlation- and concordance analyses were used to evaluate the agreement between methods. Bland–Altman plots demonstrated wide limits of agreement without significant bias for longitudinal changes in fat mass (FM), FM%, fat-free mass (FFM), and FFM% (0.39 kg (CI –0.14, 0.93), –0.39% (CI –1.06,

Abbreviations: B&A, Bland–Altman; BIA, Bioelectrical impedance analysis; BMI, Body mass index; BMI-SDS, BMI standard deviation score; CI, Confidence interval; DXA, Dual-energy X-ray absorptiometry; FFM, Fat-free mass; FM, Fat mass; IQR, Interquartile range; Lin's CCC, Lin's concordance correlation coefficient; LoA, Limits of agreement; SD, Standard deviation.

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0.26), –0.21 kg (CI –0.73, 0.31), and 0.39% (CI –0.27, 1.05), respectively). Correlations between methods for changes in FM, FM%, FFM, and FFM% were 0.93, 0.83, 0.88, and 0.83, respectively. In cross-sectional assessments, BIA underestimated FM and FM% and overestimated FFM and FFM% compared to DXA. Conclusively, BIA effectively monitors longitudinal changes in body composition at a group level in a pediatric population with obesity. However, caution is needed for individual assessments.

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1. Introduction

The continuous rise in pediatric overweight and obesity has become a critical public health concern [1] due to the associated risks of reduced quality of life [2] and the development of metabolic disorders in adulthood [3–5].

While simple anthropometric measures, such as body mass index (BMI = kg/m^2), are frequently used to evaluate the degree of adiposity or obesity, it remains an estimate with no accurate information about body composition. BMI does not differentiate between fat mass (FM) and fat-free mass (FFM) [6,7] or provide information on variations in fat distribution [8]. Therefore, a simple, accurate, and efficient method for assessing body composition and its changes is of considerable value and crucial for diagnosing and monitoring childhood obesity in clinical and research settings [9,10].

Dual-energy X-ray absorptiometry (DXA) is currently the gold standard for assessing body composition in pediatric populations due to its validity and reliability [11]. DXA utilizes low-energy X-rays to measure their attenuation as they traverse body tissues, allowing for accurate measurements of bone mineral, FM, and lean tissue mass [12]. However, DXA has downsides: it is costly, unportable, time-consuming, has limited accessibility for most practitioners, and exposes patients to a low dose of radiation [13]. Consequently, there is a need for more accessible, efficient, and user-friendly techniques [13,14].

Bioelectrical impedance analysis (BIA) has become a popular alternative to DXA, as it is a simple, portable, rapid, less expensive, and easy-to-use method [10,14]. BIA measures body composition by estimating total body water and FFM based on the body's resistance to a small electrical current using a 2-compartment model and predictive equations. FM is indirectly calculated by subtracting FFM from the total body weight [15,16]. Although the BIA technology has been shown to provide acceptable results in various populations [17–20], studies evaluating the accuracy of BIA compared to DXA in children and adolescents with obesity during lifestyle interventions, with concomitant changes in body composition, are scarce, and have provided ambiguous findings.

Existing literature has reported significant correlations between DXA and BIA but has also shown that BIA underestimates FM% and overestimates FFM compared to DXA in children and adolescents with obesity [11,21–25]; however, others have obtained opposite results [7,14,26]. Testing the ability of BIA to detect changes in body composition during a lifestyle intervention among 12-year-old adolescents with obesity, Lyra et al. [21] found that only DXA could detect both FM and FFM changes. In contrast, BIA detected only FM reductions over a 3-month intervention period. Kasvis et al. obtained similar results in children aged 7 to 13 years with overweight and obesity, with BIA accurately reporting the direction of changes in body composition (i.e., FM and FFM) in response to weight loss. However, the magnitude of these changes remained inaccurate, suggesting that they resulted from differences in body fat distribution [27]. Furthermore, sex differences have been observed in the agreement between the two methods. In a study of adolescents aged 13 to 16 years with overweight and obesity, Lazzer et al. [28] found that the differences were greater in boys compared to girls.

Few studies have compared BIA with DXA in children and adolescents with obesity engaging in long-term lifestyle interventions. To the best of our knowledge, this study is the first to involve a long-term intervention lasting more than 1 year. This study aimed to assess the level of agreement between BIA and DXA for measuring body composition and its changes during a long-term lifestyle intervention in children and adolescents with obesity, hypothesizing that BIA can accurately track longitudinal changes in body composition in this population.

2. Methods and materials

2.1. *Study design*

This project is an observational cohort study based on data from a municipality-based treatment intervention for Danish children and adolescents with obesity. Data on anthropometrics have previously been published [29].

Written informed consent for participation was obtained for each participant from their parents or legal guardians before enrollment in the intervention. This intervention served as an obesity treatment and was not primarily intended as research. According to Danish legislation, this type of study does not require formal approval or registration. Additionally, a previous analysis and publication concerning the same intervention was verified by the Central Denmark Region Committees on Health Research Ethics to not require approval from the Committees, in accordance with the Consolidation Act on Research Ethics Review of Health Research Projects—Consolidation Act number 1338 of September 1, 2020 Section 14 [30].

2.2. Definition of overweight and obesity

BMI values were adjusted for age and sex by conversion to BMI-standard deviation score (BMI-SDS) using a Danish-validated reference [31]. Childhood obesity was defined as a BMI-SDS $\geq$ 2 SD [31].

2.3. Multifactorial intervention

The intervention was a family-centered multifactorial lifestyle intervention used for the treatment of children with obesity and has previously been described in detail [29]. At baseline, the children were invited to the pediatric outpatient clinic and followed once a year for a maximum of 3 years, corresponding to a maximum of four visits. They were also seen and treated by a health nurse up to eight times annually between hospital visits. At each hospital visit, anthropometric measurements (height, weight, and waist circumference) were obtained. Body composition was assessed using both DXA and BIA. At the baseline visit, each child received an individualized treatment plan.

2.4. Participants

A total of 199 children and adolescents aged 4 to 18 years with obesity participated in the intervention at the Department of Pediatrics, Randers Regional Hospital [29]. The inclusion period extended from January 1, 2014, to December 31, 2017. Participants who had their body composition assessed using both BIA and DXA within a 30-day window, at both baseline and a subsequent follow-up visit, were enrolled in this study (Fig. 1). 15 participants were excluded due to incomplete baseline assessments, either missing DXA or BIA data, while 76 were excluded for not attending a follow-up visit. Among the 108 individuals who completed both a baseline and a follow-up visit, 16 were excluded because the interval between the DXA and BIA assessments exceeded 30 days.

2.5. Anthropometric measurements

The Tanita BC-420MA was used to measure weight and was obtained with the child wearing light clothing. Height was assessed barefoot using a fixed stadiometer and was performed by a trained healthcare professional. BMI was calculated as body mass (in kilograms) divided by height (in meters) squared and converted to BMI-SDS using a validated Danish reference [31].

2.6. Body composition

2.6.1. Dual-energy X-ray absorptiometry

The DXA scans (GE Lunar iDXA 2007) were performed by a trained radiologist, with the participant lying in a supine position and wearing light clothing. Data on FM (kg), FM%, and FFM (kg) were provided by the DXA scan, while FFM% was calculated as the percentage of FFM relative to the total body weight. FFM was defined as lean tissue mass plus bone mineral content.

2.6.2. Bioelectrical impedance analysis

BIA was performed using a Tanita BC-420MA body composition analyzer according to the manufacturer's recommendations. This foot-to-foot bioelectrical impedance device utilizes a constant current source with a high-frequency current (50 kHz, 90 μA) [32]. Participants stood barefoot on the metal footpads of the platform scale, and information on sex, body type (all set as "standard"), age, and height were entered into the device. Data on FM (kg), FM%, and FFM (kg) were provided upon completion. As BIA did not directly output FFM%, it was derived by calculating the percentage of FFM relative to the total body weight.

2.7. Statistical analyses

Study data were collected and stored using REDCap (the Research Electronic Data Capture) [33,34]. Statistical analyses were performed using GraphPad Prism (version 10.0 for Mac OS X, GraphPad Software, Boston, MA, USA) and STATA (StataCorp LP, College Station, TX, USA). Normally distributed data were reported as mean $\pm$ standard deviation (SD) and compared using a two-tailed paired *t* test (two means) for paired data and an unpaired *t* test (two means) for unpaired data. Non-normally distributed data were presented as median with interquartile range (IQR) and were compared using the Mann–Whitney *U* test for unpaired data. Normality assumptions were determined using histograms and qq-plots. Children with missing data for DXA or BIA were excluded and not reported in the tables and analyses.

A Spearman correlation analysis was conducted to evaluate the correlation between the two methods [35]. Lin's concordance correlation coefficient (CCC) (ρ_c) was estimated between BIA and DXA measurements and interpreted according to McBride [36]. Bland–Altman (B&A) plots were conducted to describe the agreement between the two methods.

The agreement between BIA and DXA for cross-sectional body composition assessments was evaluated at baseline and at the end of follow-up. Differences between baseline and the end of follow-up (the last observation) were calculated for each compartment to determine the comparability of BIA and DXA in measuring longitudinal changes in body composition.

A *P*-value of $<$.05 was considered statistically significant.

A sensitivity analysis using multiple imputation was performed to evaluate the potential impact of missing data on the results. The imputation of missing values did not alter the study's findings, and the results and conclusion remained consistent and robust.

3. Results

3.1. Baseline assessments

The median number of days between the DXA and BIA measurements at baseline was 0 days (IQR: 0-7). As outlined in Table 1, the 92 participants included in this study presented a mean age of 10.5 $\pm$ 2.9 years and a mean BMI-SDS of 3.1 $\pm$ 0.8 kg/m^2, with boys having a higher BMI-SDS than girls ($P <$ .001). No significant sex differences were observed in BIA FM (19.3 kg (IQR: 14.0-28.1) vs 19.7 kg (IQR: 13.8-25.2), $P =$ .92) or

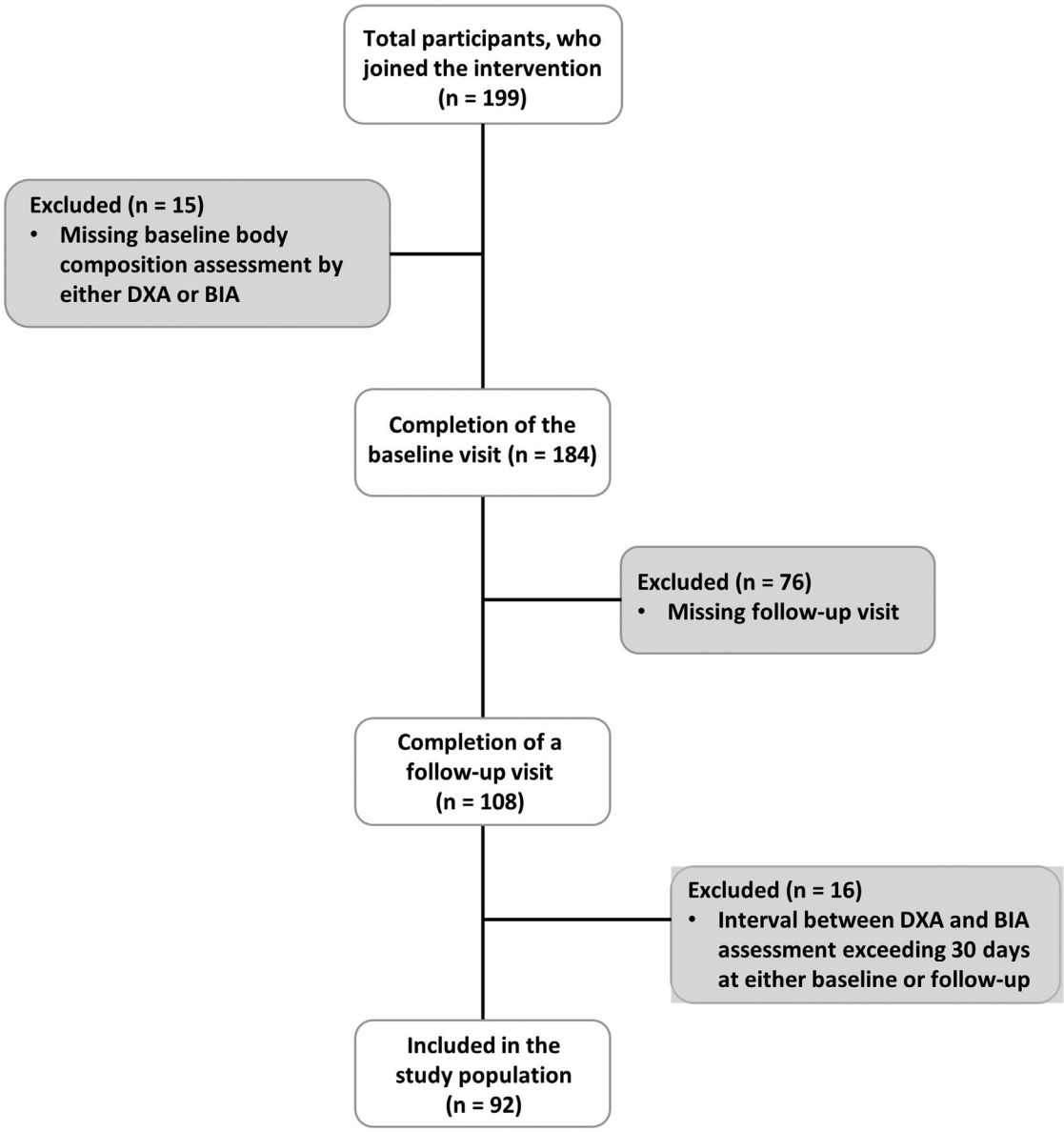

Fig. 1 – Flow chart of subject selection. DXA, dual-energy X-ray absorptiometry; BIA, bioelectrical impedance analysis.

DXA FM (27.2 ± 12.5 kg vs 24.1 ± 8.8 kg, $P = .16$), although the boys had a significantly higher BIA FFM (42.5 ± 15.2 kg vs 35.0 ± 8.9 kg, $P = .004$) and DXA FFM (36.3 ± 12.9 kg vs 31.4 ± 8.6 kg, $P = .03$), compared to the girls.

Strong correlations were found between DXA and BIA for FM (24.0 kg (IQR: 17.8-32.2) vs 19.7 kg (IQR: 14.0-27.1), $r = 0.97$, $P < .001$), FFM (32.4 kg (IQR: 24.8-39.5) vs 36.9 kg (IQR: 28.3-45.1), $r = 0.98$, $P < .001$), FM%, and FFM% (Table 2, Fig. 2E and G). The correlations between the methods, presented in Fig. 2, illustrate that BIA systematically underestimates FM% and overestimates FFM% compared to DXA. Lin's CCC showed a high level of agreement between the two methods for FM and FFM ($\rho_c = 0.89$ (95% CI 0.85, 0.92) and ($\rho_c = 0.90$ (CI 0.86, 0.92), respectively). Concordances between DXA and BIA for FM% and FFM% were poor (Table 2).

B&A plots exhibited wide limits of agreement (LoA) for all four body compartments at baseline, as seen for FM% and FFM% (Fig. 2A and C). The 95% LoA for FM and FFM were –1.21 to 9.67 kg and –10.72 to 1.01 kg, respectively. Significant bias was found in all four body compartments with a mean bias of 4.23 kg (CI 3.66, 4.80) for FM and –4.85 kg (CI –5.47, –4.23) for FFM. Differences between BIA and DXA increased with higher FM ($P = .01$), FFM ($P < .001$), and FFM% ($P < .001$, Fig. 2C), however, decreased with increasing FM% ($P < .001$, Fig. 2A).

3.2. *End of follow-up*

The median number of days between the DXA and BIA measurements at the end of follow-up was 0 days (IQR: 0-7). The mean follow-up time was 20.5 ± 9.3 months. The children sig-

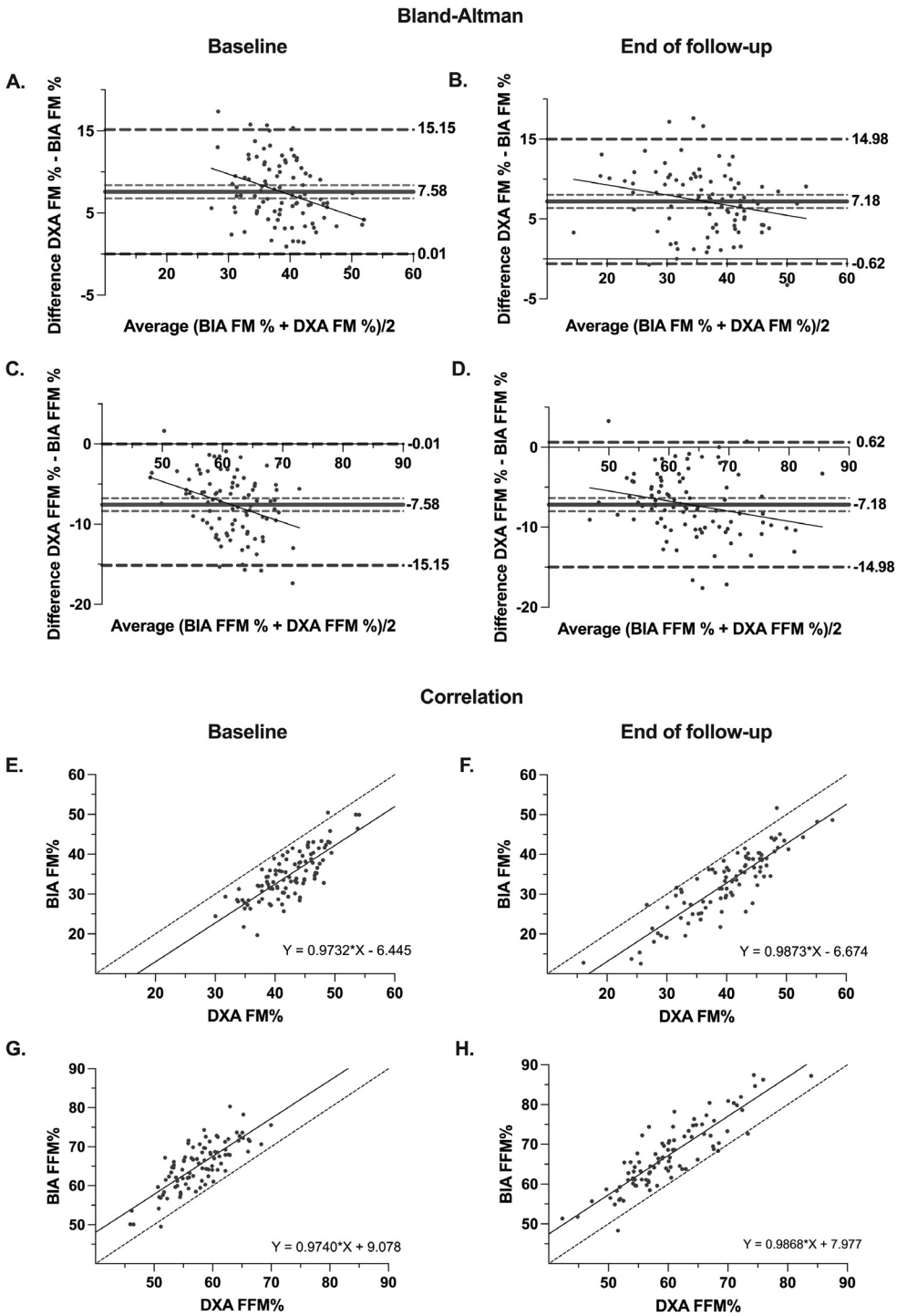

Fig. 2 – Bland–Altman- and correlation plots comparing cross-sectional measurements of FM% and FFM% at baseline and end of follow-up between BIA and DXA in 92 children and adolescents with obesity. The plots illustrate that BIA underestimates FM% and overestimates FFM% compared to DXA. Bland–Altman plots of FM% at baseline (A) and end of follow-up (B), FFM% at baseline (C) and end of follow-up (D). The solid gray line indicates the bias (mean difference), and the gray dashed lines show the 95% CI around the mean difference. The thicker gray dashed lines represent the mean ± 1.96 SD. The black solid line indicates the linear regression line of difference. Correlation plot between BIA and DXA for FM% at baseline (E) and end of follow-up (F), FFM% at baseline (G) and end of follow-up (H). The black dashed line represents the regression line in case of perfect agreement. The black solid line indicates the actual regression line derived from the comparison. BIA, bioelectrical impedance analysis; DXA, dual-energy X-ray absorptiometry; FFM, fat-free mass; FM, fat mass.

Table 1 – Baseline characteristics of children and adolescents with obesity included in the analysis, stratified into boys and girls.

	Total ($n = 92$)	Boys ($n = 45$)	Girls ($n = 47$)	P-value[a]
Age (y)	10.5 ± 2.9	10.8 ± 3.2	10.2 ± 2.5	.29
Waist circumference (cm)	90.0 ± 14.5	93.8 ± 16.1	86.3 ± 11.7	.01
Height (cm)	148.5 ± 15.8	151.1 ± 18.0	146.0 ± 13.2	.12
Weight (kg)	60.0 ± 21.6	64.3 ± 25.1	55.9 ± 16.9	.06
BMI-SDS (z-score)	3.1 ± 0.8	3.4 ± 0.8	2.8 ± 0.6	<.001
DXA-FM (%)	42.4 ± 4.9	41.7 ± 5.3	42.9 ± 4.5	.25
DXA-FFM (%)	57.7 ± 4.9	58.3 ± 5.3	57.1 ± 4.5	.25
BIA-FM (%)	34.8 ± 6.1	32.8 ± 6.7	36.7 ± 4.8	<.001
BIA-FFM (%)	65.2 ± 6.1	67.2 ± 6.7	63.4 ± 4.8	<.001

Abbreviations: BIA, bioelectrical impedance analysis; BMI, body mass index; DXA, dual-energy X-ray absorptiometry; FFM, fat-free mass; FM, fat mass; SD, standard deviation; SDS, standard deviation score.
Data presented as means ± SDs.
[a] Unpaired *t* test for comparison between boys and girls. $P < .05$ are considered significant.

Table 2 – Correlation- and concordance analyses between DXA and BIA measurements at baseline, end of follow-up, and longitudinal changes in a population of children and adolescents with obesity.

	DXA (mean ± SD)	BIA (mean ± SD)	Spearman: (r, P-value, 95% CI)	Lin's concordance coefficient: (ρ_c, 95% CI)
Baseline FM%	42.4 ± 4.9	34.8 ± 6.1	0.75, <.001, (0.64, 0.83)	0.39, (0.30, 0.48)
Baseline FFM%	57.7 ± 4.9	65.2 ± 6.1	0.75, <.001, (0.64, 0.83)	0.39, (0.30, 0.48)
End of follow-up FM%	40.0 ± 7.3	32.8 ± 8.3	0.86, <.001, (0.79, 0.91)	0.61, (0.52, 0.69)
End of follow-up FFM%	60.0 ± 7.3	67.2 ± 8.3	0.86, <.001, (0.79, 0.91)	0.61, (0.52, 0.69)
ΔFM%	–2.4 ± 5.7	–2.0 ± 5.5	0.83, <.001, (0.76, 0.89)	0.84, (0.76, 0.89)
ΔFFM%	2.4 ± 5.7	2.0 ± 5.5	0.83, <.001, (0.76, 0.89)	0.84, (0.76, 0.89)

Abbreviations: BIA, bioelectrical impedance analysis; CI, confidence interval; DXA, dual-energy X-ray absorptiometry; FFM, fat-free mass; FM, fat mass; SD, standard deviation.
$n = 92$ children and adolescents with obesity. Δ: delta-values represent the change from baseline to the end of follow-up.

Table 3 – Follow-up values of changes in age, anthropometrics, and body composition after a mean follow-up duration of 20.5 ± 9.3 months in children and adolescents with obesity.

	Difference	P-value[a]
Age (y)	1.7 ± 0.8	<.001
Weight (kg)	7.5 ± 8.2	<.001
BMI-SDS (z-score)	–0.4 ± 0.5	<.001
DXA FM (%)	–2.4 ± 5.7	<.001
DXA FFM (%)	2.3 ± 5.7	<.001
BIA FM (%)	–2.0 ± 5.5	.001
BIA FFM (%)	2.0 ± 5.5	.001

Abbreviations: BIA, bioelectrical impedance analysis; BMI, body mass index; DXA, dual-energy X-ray absorptiometry; FFM, fat-free mass; FM, fat mass; SD, standard deviation; SDS, standard deviation score.
Data presented as means ± SDs. $n = 92$ children and adolescents with obesity.
[a] Paired *t* test assessed the difference between baseline and end of follow-up. $P < .05$ are considered significant.

nificantly reduced their BMI-SDS by –0.4 ± 0.5 kg/m^2 (Table 3). Despite a significant mean increase in DXA and BIA FM (1.7 ± 6.3 kg, $P = .006$ and 1.3 ± 6.1 kg, $P = .03$, respectively), the mean DXA and BIA FM% decreased (Table 3).

Strong correlations between the methods were present for FM (24.6 kg (IQR: 18.8-31.4) vs 21.1 kg (IQR: 14.5-26.5), $r = 0.93$, $P < .001$), FFM (37.8 kg (IQR: 29.2-47.8) vs 41.9 kg (IQR: 33.4-51.2), $r = 0.98$, $P < .001$), FM%, and FFM% (Table 2, Fig. 2F and H). Fig. 2F and H illustrate a systematic bias between the methods, as BIA consistently yields lower estimates for FM% and higher estimates for FFM% than DXA. Lin's CCC showed a high level of agreement between DXA and BIA for FM and FFM ($\rho_c = 0.89$ (CI 0.85, 0.92) and $\rho_c = 0.90$ (CI 0.87, 0.93), respectively). Poor agreement was seen between the methods for FM% and FFM% (Table 2).

B&A plots showed wide LoA for all four body compartments (Fig. 2B and D for FM% and FFM%). The LoA for FM and FFM were –2.26 to 11.50 kg and –12.15 to 2.03 kg, respectively. Only minimal differences in mean biases were seen compared to baseline, with a significant mean bias of 4.62 kg (CI 3.90, 5.35) for FM and –5.06 kg (CI –5.81, –4.31) for FFM. Differences between BIA and DXA increased with higher FFM ($P < .001$) and FFM% ($P = .02$, Fig. 2D); however, decreased with increasing FM% ($P = .02$, Fig. 2B).

3.3. *Longitudinal changes*

Strong correlations were found between DXA and BIA for changes in FM (1.7 ± 6.3 kg vs 1.3 ± 6.1 kg, $r = 0.93$, $P < .001$), FFM (6.0 ± 4.8 kg vs 6.2 ± 5.3 kg, $r = 0.88$, $P < .001$), FM%, and

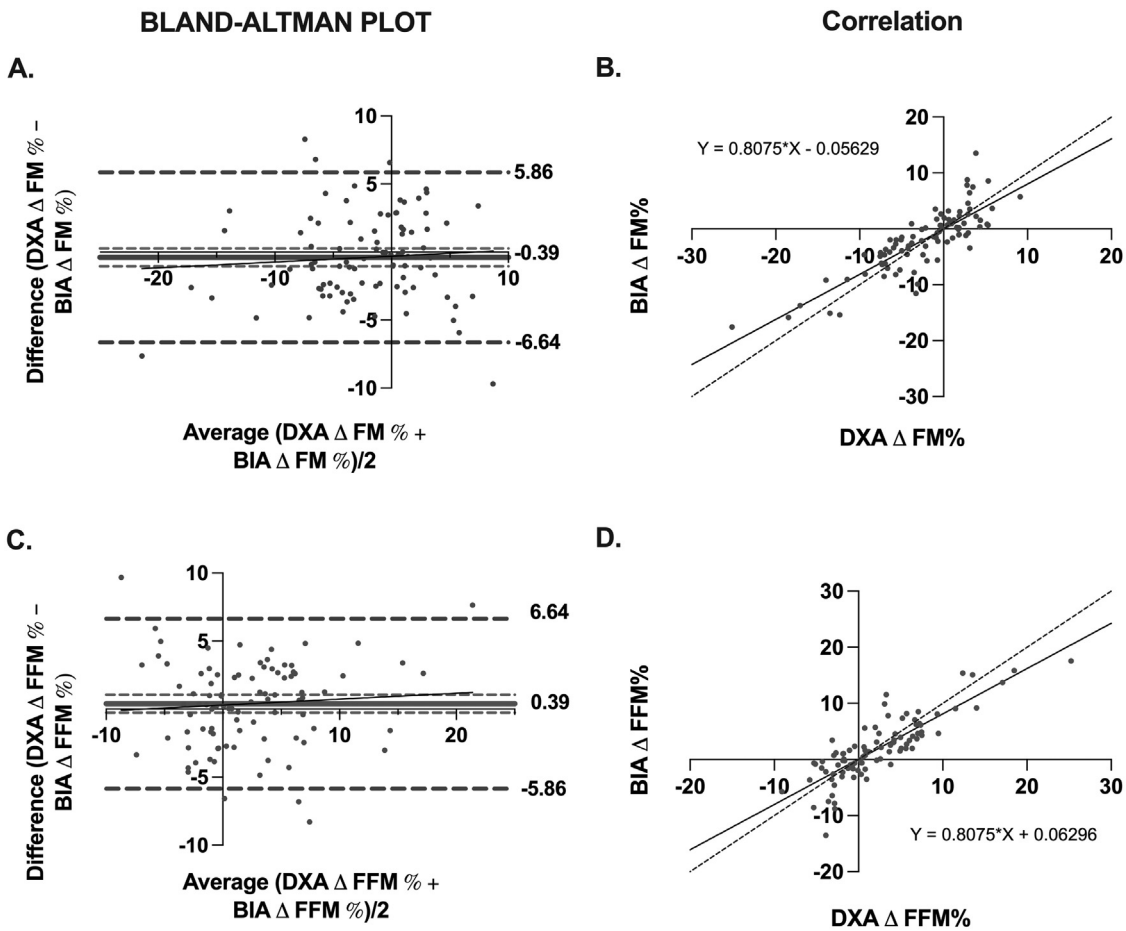

Fig. 3 – Bland–Altman- and correlation plots comparing longitudinal changes in FM% and FFM% between DXA and BIA in 92 children and adolescents with obesity, demonstrating good agreement and correlation between the two methods. Bland–Altman plots of ΔFM% (A) and ΔFFM% (C). The solid gray line indicates the bias (mean difference), and the gray dashed lines show the 95% CI around the mean difference. The thicker gray dashed lines represent the mean ± 1.96 SD. The black solid line indicates the linear regression line of difference. Correlation plot between BIA and DXA for ΔFM% (B) and ΔFFM% (D). The black dashed line represents the regression line in case of perfect agreement. The black solid line indicates the actual regression line derived from the comparison. Δ: delta-values represent the change from baseline to the end of follow-up. BIA, bioelectrical impedance analysis; DXA, dual-energy X-ray absorptiometry; FFM, fat-free mass; FM, fat mass.

FFM% (Table 2, Fig. 3B and D, respectively). Lin's CCC showed a high level of agreement between the methods for changes in all four parameters with a ρ_c of 0.91 (CI 0.87, 0.94) for FM and 0.88 (CI 0.82, 0.92) for FFM.

The B&A plots showed wide LoA for the longitudinal changes in FM (–4.64 to 5.43 kg), FFM (–5.12 to 4.70 kg), FM%, and FFM% (Fig. 3A and C, respectively). No significant bias was found in any of the four body compartments, with a mean bias of 0.39 kg (CI –0.14, 0.93) for ΔFM and –0.21 kg (CI –0.73, 0.31) for ΔFFM. Differences between BIA and DXA did not increase with greater changes in any of the four body compartments (all $P > .05$).

3.4. *Comparison of method differences between sexes*

Comparing the differences between DXA and BIA between boys and girls, boys exhibited a greater difference in all four parameters at baseline and at the end of follow-up (all $P < .05$, Table 4). However, the longitudinal changes in all four body compartments during the intervention did not differ between sexes (all $P > .05$, Table 4).

4. Discussion

The current study aimed to evaluate the level of agreement between BIA and DXA in assessing body composition and its changes during a long-term lifestyle intervention in children and adolescents with obesity. Our findings demonstrate that BIA effectively monitors longitudinal changes in body composition at a group level in children and adolescents with obesity, making it an interesting alternative to DXA in future pediatric obesity research. However, at an individual level, as in a daily clinical setting, our results do not support BIA as be-

Table 4 – Comparison of differences between DXA and BIA between sexes at baseline, end of follow-up, and for longitudinal changes in a population of children and adolescents with obesity.

		Boys ($n = 45$) Difference between methods (DXA-BIA)	Girls ($n = 47$) Difference between methods (DXA-BIA)	P-value[a]
FM (%)	Baseline	8.9 ± 4.3	6.3 ± 2.9	<.001
	End of follow-up	8.9 ± 4.0	5.6 ± 3.2	<.001
	Longitudinal changes, Δ	–0.1 ± 3.9	–0.7 ± 2.3	.3
FFM (%)	Baseline	–8.9 ± 4.3	–6.3 ± 2.9	<.001
	End of follow-up	–8.9 ± 4.0	–5.6 ± 3.2	<.001
	Longitudinal changes, Δ	0.1 ± 3.9	0.7 ± 2.3	.3

Abbreviations: BIA, bioelectrical impedance analysis; DXA, dual-energy X-ray absorptiometry; FFM, fat-free mass; FM, fat mass.
Results are presented as mean ± SD. Δ: delta-values represent the change from baseline to the end of follow-up.
[a] Unpaired t test for comparison between boys and girls. $P < .05$ are considered significant.

ing equally reliable. Results from the B&A analysis demonstrated a strong agreement between the two methods with no significant bias in tracking longitudinal changes in any of the four body compartments, suggesting accuracy at a group level. Moreover, changes in body composition demonstrated a high level of correlation and concordance between BIA and DXA. However, the LoA observed in the B&A plots may be too wide for a satisfactory assessment of changes in body composition in individual subjects. At the individual level, BIA and DXA differed by 7% or less. This suggests that in some instances, BIA could potentially replace DXA as a more efficient method to monitor changes in body composition. This is especially relevant in scenarios where substantial effects on body weight and body composition are anticipated, such as in interventions involving pharmacological treatment (e.g., semaglutide) or bariatric surgery, where substantial weight loss has been demonstrated in both pediatric and adult populations [37–39]. In such cases, despite possible discrepancies between DXA and BIA measurements, the overall trend (e.g., a substantial decrease in FM) would be observable with BIA.

BIA has previously been suggested as an effective clinical tool to measure changes in body composition at a group level in adolescents with overweight or obesity, but not at an individual level [23]. Similar findings were reported by Meredith-Jones et al. [40], who demonstrated that BIA accurately estimated changes in FM, FM%, and FFM over 1 year in both children with normal weight and those with overweight. In contrast to our results, prior research highlights the lack of accuracy of BIA for tracking changes in FFM [7,21]. Lyra et al. and Thivel et al. reported that only DXA could detect modifications in both FM and FFM during a 3-month intervention period, while BIA only identified reductions in FM [7,21]. Our study, which extended over a longer intervention period of more than 1 year, demonstrated that BIA effectively monitors changes in all four body compartments. This suggests that BIA may require more substantial changes to detect them accurately, as Lyra et al. [21] reported a considerably smaller change in BMI and FFM than ours. The same applied to Thivel et al. [7] regarding FFM.

In this study, the cross-sectional assessment of body composition at baseline and end of follow-up showed that BIA consistently underestimated FM and FM%, while FFM and FFM% were overestimated compared to DXA. These results are consistent with previous research on children and adolescents with overweight and obesity [21–23]. However, other studies found opposite results [7,14,26]. Seo et al. observed that in individuals with mild to moderate obesity, BIA tended to underestimate FM and overestimate FFM. However, in patients with severe obesity, BIA overestimated both FM and FFM compared to DXA [41]. Variations among these studies may be attributed to differences in the parameters of the BIA devices. These parameters encompass the mode of current transmission, number of electrodes, type of frequency, and specific equations integrated within the devices [42,43].

Analyzing the B&A plots for the cross-sectional baseline and end-of-follow-up data reveals that the amounts of FM and FFM influence the agreement between the two methods. Specifically, discrepancies increased with rising levels of FM, FFM, and FFM% but decreased with increasing FM%. This contrasts with Verney et al.'s [14] findings, which indicated that the comparability between the methods decreased with higher FM%. Although our results showed over- or underestimated values between the methods, the analysis demonstrated a high correlation between the two methods for all four body compartments at baseline and end of follow-up. However, Lin's concordance showed a high level of agreement for FM and FFM and poor agreement for FM% and FFM% at both time points. These findings align with recent data in healthy elementary school-aged children, showing similar concordance and correlation coefficients [44].

Given these findings, BIA is not a suitable substitute for DXA for cross-sectional assessment of body composition in children and adolescents with obesity, neither at a group nor individual level.

Our study revealed greater differences between the two methods in boys compared to girls, consistent with other studies [28]. This is likely due to differences in fat distribution between sexes during childhood. A study of 1009 individuals (aged 5-29 years) found that prepubertal girls (Tanner stage

1) had 2% more extremity fat and 6% more hip fat than boys [45]. While visceral fat accumulation appears similar in prepubertal boys and girls, girls exhibit significantly higher subcutaneous fat levels [46]. The findings from this study suggest that the android/gynoid fat distribution in this age group may be substantial enough to cause inaccuracies in BIA measurements, which could be an interesting aspect to explore further in future studies.

A few limitations of this study should be considered. Firstly, our findings only apply to the Tanita BC-420MA body composition analyzer and its prediction algorithms. As a result, generalizing these findings to the BIA method overall may be challenging [43]. Since the intervention ended, BIA technology has undergone several advancements, enhancing its accuracy and applicability. Traditionally, BIA utilized single frequency to estimate body composition. Now, devices that measure at multiple fixed frequencies (multifrequency BIA) and across a range of frequencies (bioimpedance spectroscopy) are available. These advancements allow for a more detailed assessment of body water compartments, thereby improving the accuracy of FFM and FM estimates [47]. Furthermore, segmental BIA has been developed, which measures the impedance of individual body segments instead of assuming uniform impedance across the entire body. This approach enhances the accuracy, especially in individuals with uneven fat distribution or muscle mass relative to their FFM [47,48].

Another limitation is that BIA relies on reference values established for adults, posing a limitation when applied to children and adolescents [49]. BIA estimates FFM based on the body's resistance to an electrical current and assumes a constant body resistivity, representing a fixed proportion of water in FFM. However, factors such as hydration, age, and sex can influence this assumption [50]. Children and adolescents tend to have higher hydration levels compared to adults. This leads to reduced FFM density and resistivity, resulting in an overestimation of FFM and an underestimation of FM by BIA, consistent with our results [49,51].

Furthermore, the maximum 30-day interval between BIA and DXA measurements could lead to disparities, where a gap between the measurements is present due to factors such as growth, hydration status, weight fluctuations, and changes in body composition [52,53]. Early rapid weight loss, often seen in obesity interventions, could contribute to these differences [54]. Ideally, performing all body composition measurements consecutively using both methods on the same day would have been optimal to minimize potential biases and discrepancies [52]. Participants were not required to fast before their body composition assessments, which could affect impedance, though errors are considered small [55].

In this study, we did not control for factors such as age, sex, FM, and FFM due to the limitations of the chosen statistical methods. Furthermore, the study focused on directly comparing BIA and DXA measurements without the added complexity of adjusting for these variables. However, not accounting for these factors represents a limitation, as they can influence body composition assessments and vary across different demographic groups. This limitation may lead to potential misinterpretation of the data and affect our findings' generalizability, particularly in more diverse populations.

The notable strengths of this study are the long follow-up period and the considerable homogeneity within the cohort of children and adolescents, as well as the intervention itself, which involved a small team of healthcare professionals.

A recent study reported trends in childhood BMI between 1936 and 2011, advocating for the importance of monitoring both ends of the BMI spectrum in childhood, focusing on the consequences of malnutrition and obesity [56]. It would be interesting to monitor more exact measures of body composition in future studies, considering that BMI is merely an estimate of the degree of adiposity. This highlights the importance and relevance of research like ours, which aims to validate alternative methods for monitoring more than just BMI.

5. Conclusion

In conclusion, our study demonstrates that BIA is a reliable method for monitoring longitudinal changes in body composition at a group level among children and adolescents with obesity, demonstrating strong agreement and correlation with DXA. Thus, BIA might be a feasible and efficient alternative to DXA in future pediatric obesity research. However, caution is needed if using BIA for individual assessments, as in a clinical setting, due to wide LoA, indicating potential variability when compared to DXA. Nonetheless, BIA could replace DXA in situations where substantial body composition changes are anticipated. In cross-sectional assessment of body composition, BIA consistently underestimates FM and FM% while overestimating FFM and FFM% compared to DXA. This suggests that BIA is not a suitable replacement for DXA in assessing body composition in children and adolescents with obesity, neither at a group nor individual level.

Author declarations

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CRediT authorship contribution statement

Camilla Raaby Benjaminsen: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Rasmus Møller Jørgensen:** Writing – review & editing, Supervision, Data curation, Conceptualization. **Esben Thyssen Vestergaard:** Writing – review & editing, Conceptualization. **Jens Meldgaard Bruun:** Writing – review & editing, Supervision, Conceptualization.

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Data availability statement

The authors will provide the raw anonymized data supporting this article's conclusions without undue reservation to the extent possible, following regulations and guidelines.

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REFERENCES

[1] Malecka-Tendera E, Mazur A. Childhood obesity: a pandemic of the twenty-first century. Int J Obes 2006;30:S1–3. doi:10.1038/sj.ijo.0803367.

[2] Schwimmer JB, Burwinkle TM, Varni JW. Health-related quality of life of severely obese children and adolescents. JAMA 2003;289:1813–19. doi:10.1001/jama.289.14.1813.

[3] Steinberger J, Moran A, Hong C-P, Jacobs DR, Sinaiko AR. Adiposity in childhood predicts obesity and insulin resistance in young adulthood. J Pediatr 2001;138:469–73. doi:10.1067/mpd.2001.112658.

[4] Srinivasan SR, Bao W, Wattigney WA, Berenson GS. Adolescent overweight is associated with adult overweight and related multiple cardiovascular risk factors: the Bogalusa Heart Study. Metabolism 1996;45:235–40. doi:10.1016/s0026-0495(96)90060-8.

[5] Javed A, Jumean M, Murad MH, Okorodudu D, Kumar S, Somers VK, et al. Diagnostic performance of body mass index to identify obesity as defined by body adiposity in children and adolescents: a systematic review and meta-analysis. Pediatr Obes 2015;10:234–44. doi:10.1111/ijpo.242.

[6] Wells JCK, Fewtrell MS. Measuring body composition. Arch Dis Child 2006;91:612. doi:10.1136/adc.2005.085522.

[7] Thivel D, Verney J, Miguet M, Masurier J, Cardenoux C, Lambert C, et al. The accuracy of bioelectrical impedance to track body composition changes depends on the degree of obesity in adolescents with obesity. Nutr Res 2018;54:60–8. doi:10.1016/j.nutres.2018.04.001.

[8] Luque V, Closa-Monasterolo R, Rubio-Torrents C, Zaragoza-Jordana M, Ferré N, Gispert-Llauradó M, et al. Bioimpedance in 7-year-old children: validation by dual X-ray absorptiometry—part 1: Assessment of whole body composition. Ann Nutr Metab 2014;64:113–21. doi:10.1159/000356450.

[9] Sung RY, Lau P, Yu CW, Lam PK, Nelson EA. Measurement of body fat using leg to leg bioimpedance. Arch Dis Child 2001;85:263–7. doi:10.1136/adc.85.3.263.

[10] Thomson R, Brinkworth GD, Buckley JD, Noakes M, Clifton PM. Good agreement between bioelectrical impedance and dual-energy X-ray absorptiometry for estimating changes in body composition during weight loss in overweight young women. Clin Nutr 2007;26:771–7. doi:10.1016/j.clnu.2007.08.003.

[11] Samouda H, Langlet J. Body fat assessment in youth with overweight or obesity by an automated bioelectrical impedance analysis device, in comparison with the dual-energy X-ray absorptiometry: a cross sectional study. BMC Endocr Disord 2022;22:195. doi:10.1186/s12902-022-01111-6.

[12] Marra M, Sammarco R, De Lorenzo A, Iellamo F, Siervo M, Pietrobelli A, et al. Assessment of body composition in health and disease using bioelectrical impedance analysis (BIA) and dual energy X-ray absorptiometry (DXA): a critical overview. Contrast Media Mol Imaging 2019;2019:3548284. doi:10.1155/2019/3548284.

[13] Vermeiren E, Ysebaert M, Van Hoorenbeeck K, Bruyndonckx L, Van Dessel K, Van Helvoirt M, et al. Comparison of bioimpedance spectroscopy and dual energy X-ray absorptiometry for assessing body composition changes in obese children during weight loss. Eur J Clin Nutr 2021;75:73–84. doi:10.1038/s41430-020-00738-9.

[14] Verney J, Metz L, Chaplais E, Cardenoux C, Pereira B, Thivel D. Bioelectrical impedance is an accurate method to assess body composition in obese but not severely obese adolescents. Nutr Res 2016;36:663–70. doi:10.1016/j.nutres.2016.04.003.

[15] Abbaspour A, Reed KK, Hübel C, Bulik-Sullivan EC, Tang Q, Bulik CM, et al. Comparison of dual-energy X-ray absorptiometry and bioelectrical impedance analysis in the assessment of body composition in women with anorexia nervosa upon admission and discharge from an inpatient specialist unit. Int J Environ Res Public Health 2021;18:11388. doi:10.3390/ijerph182111388.

[16] Kyle UG, Bosaeus I, De Lorenzo AD, Deurenberg P, Elia M, Gómez JM, et al. Bioelectrical impedance analysis—part I: Review of principles and methods. Clin Nutr 2004;23:1226–43. doi:10.1016/j.clnu.2004.06.004.

[17] Wang J-G, Zhang Y, Chen H-E, Li Y, Cheng X-G, Xu L, et al. Comparison of two bioelectrical impedance analysis devices with dual energy X-ray absorptiometry and magnetic resonance imaging in the estimation of body composition. J Strength Cond Res 2013;27:236–43. doi:10.1519/JSC.0b013e31824f2040.

[18] Verney J, Schwartz C, Amiche S, Pereira B, Thivel D. Comparisons of a multi-frequency bioelectrical impedance analysis to the dual-energy X-ray absorptiometry scan in healthy young adults depending on their physical activity level. J Hum Kinet 2015;47:73–80. doi:10.1515/hukin-2015-0063.

[19] Popovic V, Zerahn B, Heaf JG. Comparison of dual energy X-ray absorptiometry and bioimpedance in assessing body composition and nutrition in peritoneal dialysis patients. J Ren Nutr 2017;27:355–63. doi:10.1053/j.jrn.2017.03.003.

[20] Barreira TV, Staiano AE, Katzmarzyk PT. Validity assessment of a portable bioimpedance scale to estimate body fat percentage in white and African-American children and adolescents. Pediatr Obes 2013;8:e29–32. doi:10.1111/j.2047-6310.2012.00122.x.

[21] Lyra A, Bonfitto AJ, Barbosa VLP, Bezerra AC, Longui CA, Monte O, et al. Comparison of methods for the measurement of body composition in overweight and obese Brazilian children and adolescents before and after a lifestyle modification program. Ann Nutr Metab 2015;66:26–30. doi:10.1159/000369359.

[22] Shaikh MG, Crabtree NJ, Shaw NJ, Kirk JMW. Body fat estimation using bioelectrical impedance. Horm Res Paediatr 2007;68:8–10. doi:10.1159/000098481.
[23] Wan CS, Ward LC, Halim J, Gow ML, Ho M, Briody JN, et al. Bioelectrical impedance analysis to estimate body composition, and change in adiposity, in overweight and obese adolescents: comparison with dual-energy X-ray absorptiometry. BMC Pediatr 2014;14:249. doi:10.1186/1471-2431-14-249.
[24] Lopez-Gonzalez D, Wells JCK, Clark P. Body composition assessment in Mexican children and adolescents. Part 2: Cross-validation of three bio-electrical impedance methods against dual X-ray absorptiometry for total-body and regional body composition. Nutrients 2022;14:965. doi:10.3390/nu14050965.
[25] Huang Y, Wang X, Cheng H, Dong H, Shan X, Zhao X, et al. Differences in air displacement plethysmography, bioelectrical impedance analysis and dual-energy X-ray absorptiometry for estimating body composition in Chinese children and adolescents. J Paediatr Child Health 2023;59:470–9. doi:10.1111/jpc.16327.
[26] de Silva M, Hewawasam RP, Lekamwasam S. Concordance between body composition indices measured with dual-energy X-ray absorptiometry and bioelectrical impedance analysis in obese children in Sri Lanka. Int J Pediatr 2021;2021:6638057. doi:10.1155/2021/6638057.
[27] Kasvis P, Cohen TR, Loiselle S-È, Kim N, Hazell TJ, Vanstone CA, et al. Foot-to-foot bioelectrical impedance accurately tracks direction of adiposity change in overweight and obese 7- to 13-year-old children. Nutr Res 2015;35:206–13. doi:10.1016/j.nutres.2014.12.012.
[28] Lazzer S, Boirie Y, Meyer M, Vermorel M. Evaluation of two foot-to-foot bioelectrical impedance analysers to assess body composition in overweight and obese adolescents. Br J Nutr 2003;90:987–92. doi:10.1079/BJN2003983.
[29] Jørgensen RM, Bruun JM, Kremke B, Bahnsen RF, Nielsen BW, Vestergaard ET. Sustainable weight loss over three years in children with obesity: a pragmatic family-centered lifestyle intervention. Eat Weight Disord 2021;26:537–45. doi:10.1007/s40519-020-00887-7.
[30] Jørgensen RM, Vestergaard ET, Kremke B, Bahnsen RF, Nielsen BW, Bruun JM. The association between weight loss and long term development in quality-of-life among children living with obesity: a pragmatic descriptive intervention study. Ital J Pediatr 2022;48:135. doi:10.1186/s13052-022-01326-2.
[31] Nysom K, Mølgaard C, Hutchings B, Fleischer Michaelsen K. Body mass index of 0 to 45-y-old Danes: reference values and comparison with published European reference values. Int J Obes 2001;25:177–84. doi:10.1038/sj.ijo.0801515.
[32] Tanita.co.th. Body composition analyzer BC-420MA instruction manual, https://www.tanita.co.th/images/media/Manual_Eng_BC_420_P.pdf; [accessed 19 March 2024].
[33] Harris PA, Taylor R, Minor BL, Elliott V, Fernandez M, O'Neal L, et al. The REDCap consortium: building an international community of software platform partners. J Biomed Inform 2019;95:103208. doi:10.1016/j.jbi.2019.103208.
[34] Harris PA, Taylor R, Thielke R, Payne J, Gonzalez N, Conde JG. Research electronic data capture (REDCap)—a metadata-driven methodology and workflow process for providing translational research informatics support. J Biomed Inform 2009;42:377–81. doi:10.1016/j.jbi.2008.08.010.
[35] Schober P, Boer C, Schwarte LA. Correlation coefficients: appropriate use and interpretation. Anesth Analg 2018;126:1763–8. doi:10.1213/ane.0000000000002864.
[36] McBride G. A proposal for strength-of-agreement criteria for Lin's concordance correlation coefficient. NIWA Client Rep: HAM2005-062 2005;45:307–10.
[37] Weghuber D, Barrett T, Barrientos-Pérez M, Gies I, Hesse D, Jeppesen OK, et al. Once-weekly semaglutide in adolescents with obesity. N Engl J Med 2022;387:2245–57. doi:10.1056/NEJMoa2208601.
[38] Wilding JPH, Batterham RL, Calanna S, Van Gaal LF, McGowan BM, Rosenstock J, et al. Impact of semaglutide on body composition in adults with overweight or obesity: exploratory analysis of the STEP 1 study. J Endocr Soc 2021;5:A16–17. doi:10.1210/jendso/bvab048.030.
[39] Beamish AJ, Ryan Harper E, Järvholm K, Janson A, Olbers T. Long-term outcomes following adolescent metabolic and bariatric surgery. J Clin Endocrinol Metab 2023;108:2184–92. doi:10.1210/clinem/dgad155.
[40] Meredith-Jones KA, Williams SM, Taylor RW. Bioelectrical impedance as a measure of change in body composition in young children. Pediatr Obes 2015;10:252–9. doi:10.1111/ijpo.263.
[41] Seo Y-G, Kim JH, Kim Y, Lim H, Ju Y-S, Kang MJ, et al. Validation of body composition using bioelectrical impedance analysis in children according to the degree of obesity. Scand J Med Sci Sports 2018;28:2207–15. doi:10.1111/sms.13248.
[42] Cleary J, Daniells S, Okely AD, Batterham M, Nicholls J. Predictive validity of four bioelectrical impedance equations in determining percent fat mass in overweight and obese children. J Am Diet Assoc 2008;108:136–9. doi:10.1016/j.jada.2007.10.004.
[43] Wang L, Hui SS-c. Validity of four commercial bioelectrical impedance scales in measuring body fat among Chinese children and adolescents. BioMed Res Int 2015;2015:614858. doi:10.1155/2015/614858.
[44] Parker H, Hunt ET, Brazendale K, Klinggraeff LV, Jones A, Burkart S, et al. Accuracy and precision of opportunistic measures of body composition from the Tanita DC-430U. Child Obes 2023;19:470–8. doi:10.1089/chi.2022.0084.
[45] Taylor RW, Grant AM, Williams SM, Goulding A. Sex differences in regional body fat distribution from pre- to postpuberty. Obes 2010;18:1410–16. doi:10.1038/oby.2009.399.
[46] Arfai K, Pitukcheewanont PD, Goran MI, Tavare CJ, Heller L, Gilsanz V. Bone, muscle, and fat: sex-related differences in prepubertal children. Radiol 2002;224:338–44. doi:10.1148/radiol.2242011369.
[47] Ward LC. Bioelectrical impedance analysis for body composition assessment: reflections on accuracy, clinical utility, and standardisation. Eur J Clin Nutr 2019;73:194–9. doi:10.1038/s41430-018-0335-3.
[48] Ward LC. Segmental bioelectrical impedance analysis: an update. Curr Opin Clin Nutr Metab Care 2012;15:424–9. doi:10.1097/MCO.0b013e328356b944.
[49] Kyle UG, Earthman CP, Pichard C, Coss-Bu JA. Body composition during growth in children: limitations and perspectives of bioelectrical impedance analysis. Eur J of Clin Nutr 2015;69:1298–305. doi:10.1038/ejcn.2015.86.
[50] Lee SY, Gallagher D. Assessment methods in human body composition. Curr Opin Clin Nutr Metab Care 2008;11:566–72. doi:10.1097/MCO.0b013e32830b5f23.
[51] Fomon SJ, Haschke F, Ziegler EE, Nelson SE. Body composition of reference children from birth to age 10 years. Am J Clin Nutr 1982;35:1169–75. doi:10.1093/ajcn/35.5.1169.
[52] National Institutes of HealthBioelectrical impedance analysis in body composition measurement: NIH Technology Assessment Conference Statement. Am J Clin Nutr 1996;64 524S–532S. doi:10.1093/ajcn/64.3.524S.
[53] Chung S. Growth and puberty in obese children and implications of body composition. J Obes Metab Syndr 2017;26:243–50. doi:10.7570/jomes.2017.26.4.243.

[54] Hall KD, Kahan S. Maintenance of lost weight and long-term management of obesity. Med Clin North Am 2018;102:183–97. doi:10.1016/j.mcna.2017.08.012.
[55] Kyle UG, Bosaeus I, De Lorenzo AD, Deurenberg P, Elia M, Manuel Gómez J, et al. Bioelectrical impedance analysis—part II: utilization in clinical practice. Clin Nutr 2004;23:1430–53. doi:10.1016/j.clnu.2004.09.012.
[56] Aarestrup J, Pedersen DC, Bjerregaard LG, Jensen BW, Leth-Møller KB, Jacobsen RK, et al. Trends in childhood body mass index between 1936 and 2011 showed that underweight remained more common than obesity among 398 970 Danish school children. Acta Paediatr 2024;113:818–26. doi:10.1111/apa.16980.